

300 b/s

Bell 103 modem (1962):
two-tone FSK

33.6 kb/s

V.34 (1990s): within a few dB
of the phone line's capacity

4096

points in a Wi-Fi 7 or
DOCSIS 3.1 constellation

−1.59 dB

Shannon's limit on E_b/N_0 :
no system can go lower

3 dB

the price of orthogonal
rather than antipodal signals

6 dB

extra E_s/N_0 per extra bit
per dimension

1.53 dB

the most that constellation
shaping can ever recover

34 dB

extra SNR BPSK needs at 10^{-5}
in Rayleigh fading